www.tntextbooks.in

system is defined as "The heat absorbed by one kilogram of a substance to raise its temperature by one Kelvin at a specified temperature".

The heat capacity for 1 mole of substance, is called molar heat capacity (c_m). It is defined as "The amount of heat absorbed by one mole of the substance to raise its temperature by 1 kelvin".

Units of Heat Capacity:

The SI unit of molar heat capacity is $JK^{-1}mol^{-1}$

The molar heat capacities can be expressed either at constant volume (C_v) or at constant pressure (C_p).

According to the first law of thermodynamics

$U = q + w$ or $U = q - PdV$

$q = U + PdV$ ----------(7.19)

Differentiate (7.19) with respect to temperature at constant volume i.e dV=0,

$$\left(\frac{\partial q}{\partial T}\right)_V = \left(\frac{\partial U}{\partial T}\right)_V$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V \quad -------7.20$$

Thus the heat capacity at constant volume (C_V) is defined as the rate of change of internal energy with respect to temperature at constant volume.

Similarly the molar heat capacity at constant pressure (C_p) can be defined as the rate of change of enthalpy with respect to temperature at constant pressure.

$$c_p = \left(\frac{\partial H}{\partial T}\right)_p \quad -------(7.21)$$

Relation between Cp and Cv for an ideal gas.

From the definition of enthalpy

$H=U+PV$ ------(7.8)

for 1 mole of an ideal gas

$PV = nRT$ -----(7.22)

By substituting (7.22) in (7.8)

$H = U + nRT$ ---- (7.23)

Differentiating the above equation with respect to T,

$$\frac{\partial H}{\partial T} = \frac{\partial U}{\partial T} + nR\frac{\partial T}{\partial T}$$

$C_p = C_v + nR$ (1)

$C_p - C_v = nR$ .....(7.24)

$$\left[\because \left(\frac{\partial H}{\partial T}\right)_P = C_P \text{ and } \left(\frac{\partial U}{\partial T}\right)_V = C_V\right]$$

At constant pressure processes, a system has to do work against the surroundings. Hence, the system would require more heat to effect a given temperature rise than at constant volume, so C_p is always greater than C_v

Calculation of ΔU and ΔH

For one mole of an ideal gas, we have